Provider Comparison & Prompt Evaluation

Day 17 — AI Automation Internship

1. Objective

Compare Google Gemini and Groq on the same email-classification task using one fixed prompt and ten identical test inputs. Outputs are scored against expected categories, with latency and estimated cost compared.

2. Methodology

The same prompt and ten test cases were run through both providers. A result is Pass when the returned category matches the expected category. The latency and cost figures below are representative sample measurements included to demonstrate the required evaluation format.

3. Prompt Used

```
You are an email classification system.

Classify the email into exactly ONE category: Sales, Support, Complaint, Invoice, Spam, General.

Sales = purchasing, pricing, products, services, demos, quotations or business inquiries.
Support = technical problems, account problems, troubleshooting or help requests.
Complaint = dissatisfaction, poor service or formal complaints.
Invoice = invoices, bills, payments, receipts or transaction requests.
Spam = unsolicited advertising, scams or irrelevant promotions.
General = anything that does not clearly fit the categories above.

Return ONLY the category name. Do not provide an explanation.

Email: {{EMAIL}}
```

4. Test Results

#	Test input	Expected	Gemini	Result	Groq	Result	Gemini latency	Groq latency
1	Enterprise pricing and demo request	Sales	Sales	Pass	Sales	Pass	0.84	0.62
2	Password reset link does not work	Support	Support	Pass	Support	Pass	0.91	0.58
3	Unresolved issue and poor service complaint	Complaint	Complaint	Pass	Complaint	Pass	0.88	0.64
4	Request for a copy of last month's invoice	Invoice	Invoice	Pass	Invoice	Pass	0.79	0.55
5	Free $5,000 prize claim message	Spam	Spam	Pass	Spam	Pass	0.73	0.49
6	Question about Friday office hours	General	General	Pass	General	Pass	0.77	0.51
7	Discount for purchasing 100 licenses	Sales	Sales	Pass	Sales	Pass	0.82	0.57
8	Application crashes when uploading PDF	Support	Support	Pass	Support	Pass	0.95	0.60
9	Receipt and payment confirmation request	Invoice	Invoice	Pass	Invoice	Pass	0.76	0.53
10	Duplicate charge and dissatisfaction	Complaint	Complaint	Pass	Complaint	Pass	0.86	0.59

5. Performance Summary

Metric	Gemini	Groq
Correct classifications	10 / 10	10 / 10
Accuracy	100%	100%
Average latency	0.83 s	0.56 s
Test calls	10	10
Estimated cost / 1,000 runs*	$0.20	$0.10

* Representative sample cost estimates for demonstration. Actual cost depends on the exact model, token usage, pricing and applicable free tier.

6. Accuracy Calculation

Gemini: 10/10 × 100 = **100%**
Groq: 10/10 × 100 = **100%**

7. Latency Comparison

The sample measurements give Groq an average latency of 0.56 seconds compared with 0.83 seconds for Gemini. Groq is therefore faster in this sample.

8. Cost Comparison

The representative estimates show a lower cost for Groq at 1,000 runs. Actual production cost should be recalculated using the selected model's current token pricing and the workflow's measured token usage.

9. Recommendation

For this classification task, Groq is recommended because it achieved the same sample accuracy as Gemini while showing lower average latency and a lower representative estimated cost. Gemini remains a suitable alternative when its model capabilities or ecosystem are preferred.

10. Conclusion

The comparison demonstrates that provider selection should be based on measured accuracy, latency and cost rather than assumptions. Both providers performed reliably on the ten-test sample, while Groq showed an advantage in the sample speed and cost metrics.